An Infrared Small Target Detection Method for Remote Sensing Images Based on Improved RT-DETR

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***Abstract*—Due to the geometric characteristics of remote imaging and the interference of background clutter, infrared targets usually exhibit features such as small size, low contrast, and low signal-to-noise ratio, which greatly increases the difficulty of target detection task in complex environments. Therefore, this paper proposes a remote sensing image infrared small target detection method RIST-RTDETR based on RT-DETR. This method designs a Hybrid Frequency-Spatial feature extraction module, which employs a feature concatenation and adaptive fusion strategy. It aligns and integrates frequency-domain enhanced features based on spatial features, effectively suppressing feature shifts. Additionally, in the neck network, the Haar Wavelet Downsampling module is used to replace the traditional downsampling operation, to enhance high-frequency information and suppress background noise. At the same time, the p2 detection layer is introduced to further improve the detection accuracy of small targets. The experimental results on HIT-UAV dataset show that compared with the baseline model RT-DETR, RIST-RTDETR achieves a mAP50 of 94.6% while reducing parameters by 25.6%. This fully validates the effectiveness of this method in remote sensing infrared small target detection.**

Keywords—Infrared Small Target, Deep learning, Object Detection, Remote Sensing Images, RTDETR

I. INTRODUCTION

Infrared small target detection is one of the important research directions in intelligent processing of remote sensing images. It mainly obtains the thermal radiation difference between target and background through infrared imaging equipment and realizes the positioning and recognition of weak and small targets by means of image processing or target detection algorithms [1]. Compared with visible imaging, infrared imaging is less affected by lighting conditions, has strong day-and-night working ability, good concealment, and strong anti-interference ability, etc. Therefore, it has wide application value in military reconnaissance, sea surface surveillance, forest fire prevention, and emergency search and rescue, etc. [2,3]. However, in actual remote sensing scenarios, infrared small targets usually occupy only a very small area of pixels in image, lack obvious texture, shape and edge information, and are also easily affected by complex ground object backgrounds, cloud layer noise, sensor noise and weak contrast imaging conditions. These factors make infrared small target detection still a challenging task.

The current infrared small target detection methods can be broadly classified into traditional methods and those based on deep learning. Traditional methods mostly rely on filtering, local contrast, background suppression, and low-rank sparse decomposition strategies, enhancing the difference between target and background through artificially designed features [4]. These methods can achieve certain results under conditions where background is relatively simple and target features are relatively stable. However, their detection performance often depends on prior assumptions and parameter settings. When dealing with complex remote sensing backgrounds, significant changes in target size, or strong noise interference, traditional methods tend to have high false alarm rates, serious missed detections, and insufficient generalization ability. Moreover, as the resolution of remote sensing images continues to increase, the scale of the data to be processed significantly expands. Traditional algorithms gradually struggle to meet the practical application requirements in terms of computational efficiency and real-time performance.

In contrast, deep neural networks can automatically learn discriminative semantic information and multi-scale features from large-scale samples, thereby improving detection robustness in complex backgrounds. Among them, Transformer -based detection frameworks have attracted increasing attention due to their strong global context modeling capability. DETR (Detection Transformer) [5] first introduced Transformer into end-to-end object detection, achieving localization and

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classification through set prediction while avoiding anchor design and non-maximum suppression. However, its slow convergence and high computational cost limit its use in real-time scenarios.

To solve this problem, RT-DETR [6] was proposed to retain the advantages of end-to-end detection while optimizing the network structure and query mechanism, achieving a good balance between detection speed and accuracy, and demonstrating better detection performance than some YOLO series models at the same scale. Nevertheless, directly applying RT-DETR to remote sensing infrared small target detection remains insufficient. On the one hand, the features of small targets are easily weakened or lost during the deep propagation process; on the other hand, the background of remote sensing infrared images is complex, and the model needs stronger local detail perception ability and multi-scale feature fusion ability; in addition, the large model parameters and computational cost also limit its application on edge devices or real-time monitoring platforms.

In response to the aforementioned issues, this paper proposes a method for detecting infrared small targets in remote sensing images, which not only improves the detection accuracy but also maintains the detection speed of the model. Firstly, to address insufficient global context modeling, feature shift, and detail loss in infrared small target detection, this paper proposes a Hybrid Frequency-Spatial Module, which adopts a dual-branch design to preserve spatial details and capture global correlations. In addition, Haar Wavelet Downsampling is introduced to retain edge and texture information while reducing computational cost. Finally, a P2 detection layer is added to the neck to exploit low-level features and improve sensitivity to tiny targets.

II. METHOD

A. Algorithm Overview

As the first end-to-end detector in the DETR series to overcome the real-time bottleneck, RT-DETR employs an efficient hybrid encoder and an uncertainty-minimizing query selection mechanism, significantly reducing inference latency while eliminating the need for non-maximum suppression (NMS) post-processing. Considering the severe constraints of onboard computational resources in UAVs, this study adopts the lightweight ResNet-18 as the backbone of RT-DETR. The optimized architecture is shown in Fig. 1.

B. Hybrid Frequency-Spatial Module

In the backbone network, RIST-RTDETR proposes the hybrid frequency-spatial module (HFSM) with hierarchical feature fidelity by deeply integrating Fused Fourier Convolution Mixer (FFCM) frequency-domain processing unit [7] and Cross Stage Partial (CSP) structure [8]. Based on the branch partitioning strategy of CSP, this module divides input features into a basic branch and an enhanced branch. Specifically, given the input feature $X \in \mathbb{R}^{C*H*W}$, HFSM first splits it into two branches:

$$X_s,\ X_f = Split\left(X\right) \tag{1}$$

Among them, X_s retains the original spatial details through cross-stage connections, while X_f serves as the spatial anchor for subsequent fusion. The spatial branch preserves detailed local information:

$$F_s = \mathrm{C_s}\left(X_s\right) \tag{2}$$

while the frequency branch enhances global contextual information:

$$F_f = C_f\left(\mathrm{FFCM}\left(X_f\right)\right) \tag{3}$$

where $C_s\left(\cdot\right)$ represents the convolution transformation in the spatial branch; $FFCM\left(\cdot\right)$ represents the frequency domain feature enhancement unit; $C_f\left(\cdot\right)$ represents the convolution transformation in the frequency domain branch. F_s and F_f are the output features of the spatial branch and the frequency domain branch, respectively.

After channel alignment, the two branches are fused by a learnable fusion operation:

$$F_{out} = Conv\left(Concat\left(F_s, Conv(F_f)\right)\right) \tag{4}$$

where $Conv\left(\cdot\right)$ is used for channel alignment and feature fusion; $Concat\left(\cdot\right)$ indicates feature concatenation along the channel dimension. F_{out} r represents the fused features output by the HFSM module at the end.

This structure not only prevents the frequency-domain branch from dominating the feature distribution, but also enables efficient interaction and complementary enhancement between frequency-domain and spatial features with the supplement of spatial priors. The final fused features thus maintain both high-fidelity spatial details and rich frequency-domain semantics, which reduces information distortion.

C. Haar Wavelet Downsampling

The Haar Wavelet Downsampling (HWD) [9] module leverages wavelet transformation to decompose input feature maps into multiple scales, thereby reducing spatial resolution while preserving essential edge and texture information. Specifically, this module employs Haar wavelets to split feature maps into low-frequency and high-frequency sub-bands. The low-frequency components provide downsampled representations of global semantic information, while the high-frequency components carry edge and texture details that can assist in subsequent feature reconstruction. Compared with traditional max pooling or convolutional downsampling, Haar Wavelet Downsampling not only reduces computational and memory costs but also better preserves small targets and structural details, offering effective support for multi-scale feature fusion and fine-grained object detection.

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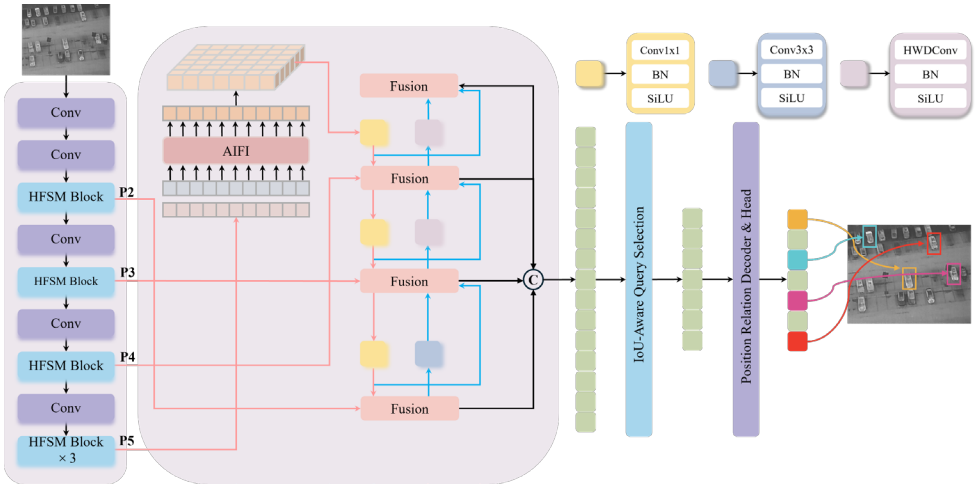


Fig. 1. Overview of RIST-RTDET.

D. P2 Detection Layer

The original RT-DETR mainly relies on relatively deep feature maps for detection. For infrared small targets in HIT-UAV, many targets occupy only a small number of pixels. After repeated downsampling, these targets may correspond to very few activation cells on high-level feature maps, which weakens localization cues and increases missed detections. Therefore, introducing the P2 layer allows the detector to exploit higher-resolution low-level features and preserve fine-grained spatial information.

III. Experiment And Analysis

A. Dataset

HIT-UAV dataset [10] is an open high-altitude UAV infrared target detection dataset with 2898 thermal infrared images. Collected by UAV under varying flight altitudes (60-130 m), shooting angles (30°-90°), scenes, weather and light conditions. Affected by UAV movement, light changes, target occlusion and complex background interference, the targets are mostly small-scale with low contrast to the surrounding environment, which fully proves the applicability and challenge of this dataset for verifying our algorithm.

B. Experiment settings

a) Evaluation index: To evaluate the performance of RIST-RTDETR, adopt several key evaluation metrics: Precision (P), Recall (R), F1 score, mean Average Precision (mAP), Parameters (Params) and FLOPs. Among them, mAP refers to the average precision across all categories (mAP50 denotes an IoU threshold of 0.5); Params quantifies the computational and storage resources consumed by the model; FLOPs represents the number of floating-point operations required for a single inference, serving as an indicator to measure model's computational efficiency.

b) Parameter setting: To assess the effectiveness of RIST-RTDETR, this paper designs ablation experiment and comparative experiment based on HIT-UAV dataset. The experimental setup, including software and hardware configurations, as well as key training parameters, are detailed in TABLE I and TABLE II, to ensure the repeatability and fairness of the experiment.

TABLE I. Training Environment Configuration

Parameters	Configuration
CPU	Intel(R) Xeon(R) Platinum 8488C
GPU	NVIDIA A100
GPU memory size	80G
Operating systems	Windows 10
Language	Python 3.10.17
Frame	Pytorch 2.1.1
CUDA Version	Cuda 11.8

TABLE II. Training Training Parameter Setting

Parameters	Setup
Epochs	300
Image size	640×512
Batch size	16
Worker	4
Learning rate	0.01
Momentum	0.9
Optimizer	AdamW

C. Ablation Experiments

To verify the effectiveness of each proposed module, we perform ablation studies based on the baseline RT-DETR on the HIT-UAV infrared dataset, and analyze the influence of each improved structure on the overall model step by step. The corresponding experimental results are summarized in TABLE III.

TABLE III. Ablation Experimental Results

Models	P (%)	R (%)	mAP50 (%)	Params (M)	FLOPs (G)
RT-DETR	90.7	88.5	93.4	19.9	57.0
RT-DETR+HFSM	91.5	89.7	94.2	16.6	50.6
RT-DETR+HWD	90.3	89.0	93.3	19.2	68.5
RT-DETR+P2	**91.7**	88.9	93.8	18.6	78.2
RT-DETR+HWD+P2	91.1	89.4	94.1	15.3	72.6
RIST-RTDETR	**91.7**	**89.9**	**94.6**	**14.8**	**55.6**

Note: Bold values indicate the best results.

Firstly, replacing the original backbone with the Hybrid Frequency-Spatial network effectively improves detection accuracy by enhancing the model's attention to detail and edge features in low-contrast environments. Secondly, adopting HWD instead of conventional downsampling accelerates the overall detection speed. Thirdly, the P2 layer strengthens the

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model's sensitivity to small targets by focusing on low-level features. Overall, the improved model achieves better efficiency in parameters and FLOPs, with mAP50 increased by 1.3% over the baseline. These results validate the effectiveness of the proposed strategies for infrared small target detection in remote sensing images.

D. Comparative Experiments

To further verify the performance of RIST-RTDETR method, this paper selects multiple benchmark algorithms, including YOLO and RT-DETR for comparison. TABLE IV summarizes the experimental results of the proposed RIST-RTDETR and the five other comparative algorithms.

As shown in TABLE IV, compared with the original RTDETR, the proposed RIST-RTDETR reduces Params by 25.6% and FLOPs by 2.5%, while the mAP50 at an IoU threshold of 0.5 reaches 94.6%. Furthermore, relative to YOLO series models such as YOLOv5m, YOLOv8m and YOLOv10m, RIST-RTDETR achieves the best results in mAP50, Params and FLOPs. They are respectively superior to the comparison models YOLOv5m (+13.2, -10.2, -8.4), YOLOv8m (+12.3, -11.0, -23.1), and YOLOv10m (+12.8, -0.5, -3.3). In addition, compared with RT-DETR-R34, mAP50 decreases by only 1.1%, while Params and FLOPs dropped by 52.4% and 37.4% respectively. These results verify the effectiveness of our method for infrared low-contrast small target detection in complex scenarios. Meanwhile, for resource-constrained embedded devices, the proposed approach can effectively alleviate the problem of excessive computation that limits the practical deployment of conventional detection models.

TABLE IV. COMPARATIVE EXPERIMENTAL RESULTS

Models	F1 (%)	mAP50 (%)	Params (M)	FLOPs (G)
YOLOv5m	77.4	81.4	25.0	64.0
YOLOv8m	78.2	82.3	25.8	78.7
YOLOv10m	77.8	81.8	15.3	58.9
RT-DETR	89.7	93.4	19.9	57.0
RT-DETR-R34	90.2	93.6	31.1	88.8
RIST-RTDETR	**91.0**	**94.6**	**14.8**	**55.6**

Note: Bold values indicate the best results.

E. Visualization Experiment

Fig.2 shows the visual comparison experiment of RT-DETR and RIST-RTDETR. The green dotted rectangle highlighting the region of interest for detailed analysis.

To more intuitively verify the effectiveness of the proposed improvement strategy, we conducted a visual experiment on the test set, as shown in Fig.2. The experiment still selected various types of detection scenarios with low contrast, overlapping density, small targets, different shooting heights and angles as detection samples to compare the detection performance of the basic model RT-DETR and RIST-RTDETR. The green dotted box in the figure represents the comparison area of this experiment. From the figure, it can be seen that the method proposed in this paper not only effectively detects the targets missed by RT-DETR, but also significantly reduces the false detection of low-contrast targets.

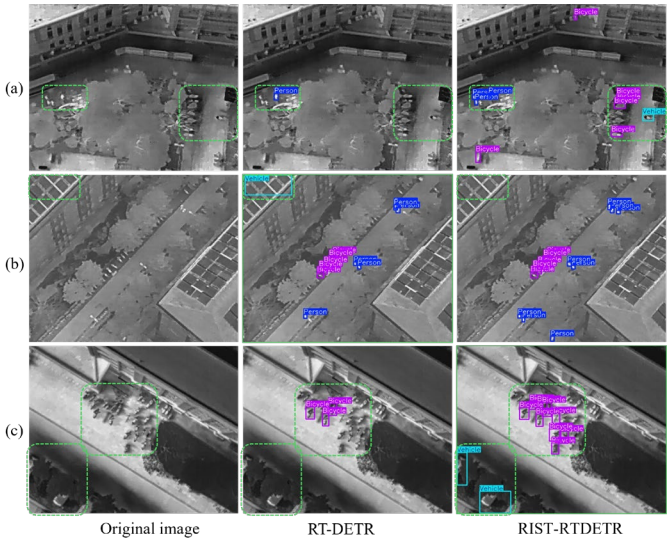


Fig. 2. Comparison results of visualization experiment conducted on the HIT-UAV dataset.

IV. CONCLUSION

This paper develops the remote sensing image infrared small target detection method RIST-RTDETR. By optimizing the backbone and the neck of RT-DETR, significantly enhancing the model's detection accuracy and efficiency. Experimental results on HIT-UAV infrared dataset demonstrate that, compared with RT-DETR, our method not only improves the precision of infrared small targets but also reduces parameters and computational complexity, providing possibility for the real-time deployment of the proposed model on UAV platform.

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